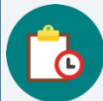


3.1 Product Laws of Exponents

Lesson Introduction

 Benchmark	MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions limited to whole-number exponents and rational-number bases.		 Learning Target	I can explain and use the power of a product, power of a power, product of powers, and identity laws of exponents.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.3.3 Evaluate positive rational numbers with natural-number exponents.		MA.8.AR.1.1 Apply the laws of exponents to generate equivalent algebraic expressions limited to integer exponents and monomial bases. MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expression, limited to integer exponents and rational-number bases, with procedural fluency.	
 Essential Questions	How can the laws of exponents be used to write an equivalent expression?			
 Lesson Narrative	In this lesson, students explore four laws of exponents: product of powers, power of a product, power of a power, and identity exponent law. Students begin by exploring each law through a series of collaborative Explorations designed to have them discover the law prior to being formally introduced to each one. The lesson concludes with students identifying the law that will be used and then using it to simplify a given expression.			
 Component Summary	3.1.1 Warm-Up (~5 min)	Reflect on growth mindset video (<i>SE p. 99</i>)	3.1.4 Exploration (5-10 min)	Power of a power exponent law exploration activity (<i>SE p. 103</i>)
	3.1.2 Exploration (5-10 min)	Product of powers exponent law exploration activity (<i>SE p. 100</i>)	3.1.5 Exploration (5-10 min)	Identity exponent law exploration activity (<i>SE p. 104</i>)
	3.1.3 Exploration (5-10 min)	Power of a product exponent law exploration activity (<i>SE p. 102</i>)	3.1.6 Your Turn! (5-10 min)	Identifying and applying product laws of exponents activity (<i>SE p. 105</i>)
	3.1.7 Wrap-Up (~5 min)	Ticket Out the Door on identifying the exponent law and equivalent expression for a given expression (<i>SE p. 105</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Base - Exponent - Product of powers exponent law - Power of a product exponent law - Power of a power exponent law - Identity exponent law <u>Additional Terms:</u> N/A		- Calculators - Growth Mindset: Mistakes Help You Grow video	(Optional) Foldable, Frayer Model, anchor chart or other guided-note scenarios. (Optional) Warm-Up video

 Teacher Tips	Common Misconceptions • Students may forget that the bases need to be the same in order to apply the product of powers law. • Students may not use the negative sign when applying the laws to negative numbers. • Students may add instead of multiply when using the power of a power law.	Things to Consider • Allow students to use a calculator to evaluate. The evaluation shows students that two numeric expressions are equivalent, which then confirms the law to the student. • If time is an issue, then consider assigning 3.1.5 Exploration and 3.1.6 Your Turn! for homework.
	Literacy and Language Routines Clarify, Critique, Correct In each Exploration activity, students will be given the opportunity to explore scenarios that will result in their discovery of a product law of exponents. When closing each activity, consider leveraging this protocol to have students critique their findings and correct based on feedback and the formal definition, when introduced.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection This lesson contains four Explorations, where students will work collaboratively with a partner to discover the four laws of exponents: product of powers, power of a product, power of a power, and identity. Through these activities, students will engage in collaborative discussion and reflect on their findings when introduced to the formal definition of each law.
	 Differentiate Instruction Consider grouping or partnering students together based on their learning styles. This lesson has four Explorations, and grouping or pairing students by their learning styles can help them progress through the activities more fluidly as well as make potential scaffolding more accessible when leveraged. Consider creating a foldable, Frayer Model, or note-taking guide to keep a central location for these specific laws of exponents. This will allow students to refer back to them as they continue to practice throughout the unit. 3.1.7 Wrap-Up activity of this lesson provides a Ticket Out the Door experience where students will identify the law of exponents that should be used and the resulting expression when it is used. This can provide data on which students may need additional support.	
Lesson Notes:		

3.1.1 Warm-Up: Mistakes Help You Grow

Component Overview: Play the *Growth Mindset: Mistakes Help You Grow* video for the entire class. After students watch the video (1:52 minutes), prompt them to reflect on the video by answering the question independently.



3.1.1 Warm-Up: Mistakes Help You Grow

- Two artists are drawing a plant. As they go through the process, both artists make mistakes and start over on their drawings. One of the artists crumples their papers up and throws them on the floor. The other artist hangs them on the wall next to where she is working.



What do you think is the purpose of hanging up the drawings with mistakes?

Answers will vary. Hanging up drawings with mistakes allows us to see drawings that need edits. It also allows us to remember that mistakes happen, and lead to growth.



How could you apply this example to your work in math class? How do you feel when you make a mistake? Why do you feel this way? Why are making mistakes beneficial?

3.1.2 Exploration: Product of Powers Exponent Law

Component Overview: Students will work collaboratively to discover the product of powers exponent law.



3.1.2 Exploration: Product of Powers Exponent Law

Recall that repeated multiplication is a way of writing a number with an exponent.

The number 2, called the **base**, is being multiplied repeatedly.

$$2^3 = 2 \times 2 \times 2$$

base → 2 is being multiplied repeatedly



A **base** is the number used as a factor in exponential form.

Since the base, 2, is a factor 3 times, the **exponent** is 3.

$$2^3 = 2 \times 2 \times 2$$

exponent → 2 is a factor 3 times



An **exponent** is the number of times the base occurs as a factor.



The language of “base” and “exponent” are not new for students at this grade level. Consider asking students to recall what they know on the concept of the base and exponent to set students on the right track prior to releasing them to complete the remainder of the activity with their partner or small group.

3.1.2 Exploration: Product of Powers Exponent Law (continued)

1. With your partner, use the definition of an exponent to complete the table.

- a. Write each product as a repeated multiplication expression. Then, write each expression with an exponent.

Product	Repeated Multiplication	Power of the Form a^b
$2^2 \cdot 2^3$	$(2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$	2^5
$3^5 \cdot 3^4$	$(3 \cdot 3 \cdot 3 \cdot 3 \cdot 3) \cdot (3 \cdot 3 \cdot 3 \cdot 3)$ $= (3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3)$	3^9
$5^1 \cdot 5^4 \cdot 5^3$	$(5) \cdot (5 \cdot 5 \cdot 5 \cdot 5) \cdot (5 \cdot 5 \cdot 5)$ $= (5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5)$	5^8

- b. How do the exponents in the first column relate to the exponents in the last column?
The sum of the exponents in the first column is equal to the exponent in the last column.
- c. What do you notice about the bases in all three columns?
Going across each row, the bases are the same from column to column.
2. Write a rule for the product of two exponential expressions with the same base.
Answers will vary. When multiplying exponents with the same base, add the exponents.



Product of Powers Exponent Law

When multiplying two powers that have the same nonzero base, add the exponents and keep the same base.

3. With your partner, follow the steps below.
- Determine which column you will start in, and which column your partner will start in.
 - Using the directions for your column, evaluate the expression. Your partner should use the directions in their column to evaluate the same expression.
 - Verify if your answers are the same. If not, find the error(s).
 - After each problem, switch columns.



Consider how to utilize each Exploration for your class. For classes that work well with their partners, consider having students complete each Exploration, checking in with you before completing the next Exploration. For classes that may struggle, consider combining whole and small group by determining which questions to pull the students together as a whole group. Ensure that students understand the law that was discovered before moving to the next Exploration.



Use inquiry to prompt exploration or encourage students to make explicit connections between their work and their glossary term, product of powers.

- How does repeated multiplication show that the exponents will be added?
- What is the benefit of using the product of powers exponent law instead of repeated multiplication?



As students begin to learn the exponent laws and their formal definitions, consider leveraging a vocabulary device to help them create a single reference point. This can be done through a foldable, a series of Frayer Models, or other guided-note scenarios. This will provide students their own reference guide when solving problems throughout this unit.

3.1.2 Exploration: Product of Powers Exponent Law (continued)

The first expression has been evaluated.

Evaluate each power, then multiply.	Apply the product of powers exponent law, then evaluate the power.
$7^2 \cdot 7^1$	
$7^2 \cdot 7^1$ $49 \cdot 7$ 343	$7^2 \cdot 7^1$ $7^{(2+1)}$ 7^3 343
$6^9 \cdot 6^3$	
$6^9 \cdot 6^3$ $10077696 \cdot 216$ $2,176,782,336$	$6^9 \cdot 6^3$ 6^{9+3} 6^{12} $2,176,782,336$
$(-8)^4 \cdot (-8)^5$	
$(-8)^4 \cdot (-8)^5$ $4096 \cdot -32768$ $-134,217,728$	$(-8)^4 \cdot (-8)^5$ $(-8)^{4+5}$ $(-8)^9$ $-134,217,728$



For the table, allow students to use a calculator to evaluate. The evaluation shows students that two numeric expressions are equivalent.



Watch for misconceptions on the third problem in the table. Students often forget to use the negative sign in their repeated multiplication or when inputting into their calculator.



Using the method used in the table, how can you verify if Julian is correct?

Why do the bases have to be the same for the law to apply?

4. Julian says that he can evaluate $3^2 \cdot 2^3$ using the product of powers exponent law.

- a. Explain to Julian if he is correct.

Julian is incorrect. The bases have to be the same for him to use the product of powers exponent law.

- b. Complete the statement about the product of powers exponent law.

When a is a nonzero real number, and m and n are

integers, the exponential expression $a^m \cdot a^n$ is

equal to

- ☐ a^{mn}
☐ a^{m-n}
☒ a^{m+n}

3.1.3 Exploration: Power of a Product Exponent Law

Component Overview: Students will work collaboratively to discover the power of a product exponent law.



3.1.3 Exploration: Power of a Product Exponent Law

- With your partner, complete the table and answer the questions.
 - Write each power of a product as a repeated multiplication expression, and group repeated factors by using the commutative and associative properties. Then, write each expression with an exponent in the form $a^m \cdot b^n$.

Power of a Product	Repeated Multiplication	Group Repeated Factors	Power of the Form $a^m \cdot b^n$
$(2 \cdot 3)^3$	$(2 \cdot 3) \cdot (2 \cdot 3) \cdot (2 \cdot 3)$	$(2 \cdot 2 \cdot 2) \cdot (3 \cdot 3 \cdot 3)$	$2^3 \cdot 3^3$
$(6 \cdot 5)^4$	$(6 \cdot 5) \cdot (6 \cdot 5) \cdot (6 \cdot 5) \cdot (6 \cdot 5)$	$(6 \cdot 6 \cdot 6 \cdot 6) \cdot (5 \cdot 5 \cdot 5 \cdot 5)$	$6^4 \cdot 5^4$
$(10 \cdot 4)^2$	$(10 \cdot 4) \cdot (10 \cdot 4)$	$(10 \cdot 10) \cdot (4 \cdot 4)$	$10^2 \cdot 4^2$

- How does the base in the first column relate to the bases in the last column?
The bases in the first and last column are the same. The bases in the first column is the product of the bases in the last column.
 - How does the exponent in the first column relate to the exponents in the last column?
The exponents in the first column are the same as the exponents in the last column, but the exponents have been distributed to each of the bases.
- Write a rule for the power of a product.
Answers will vary. When raising a product to a power, distribute the exponent to each factor.



Power of a Product Exponent Law

When raising a product to a power, distribute the power to each factor.



How did the rules that you came up with compare to the actual exponent law shown? How did grouping the repeated factors lead to the discovery of your rule?



Allow students the time and opportunity to add the formal information for the power of a product exponent law to their note-taking guide, foldable, or additional Frayer Model to maintain their working reference guide.

3.1.3 Exploration: Power of a Product Exponent Law (continued)

3. With your partner, follow the steps:
- Determine which column you will start in, and which column your partner will start in.
 - Using the directions for your column, evaluate the expression. Your partner should use the directions in their column to evaluate the same expression.
 - Verify if your answers are the same. If not, find the error(s).
 - After each problem, switch columns.

The first expression has been evaluated.

Evaluate inside the parentheses, then apply the exponent.	Apply the power of a product exponent law, then evaluate the power.
$(-5 \cdot 6)^2$	
$(-5 \cdot 6)^2$ $(-30)^2$ $(-30) \cdot (-30)$ 900	$(-5 \cdot 6)^2$ $(-5)^2 \cdot 6^2$ $25 \cdot 36$ 900
$(-6 \cdot 4)^5$	
$(-6 \cdot 4)^5$ $(-24)^5$ -7,962,624	$(-6 \cdot 4)^5$ $(-6)^5(4^5)$ $-7,776 \cdot 1,024$ -7,962,624
$(-3 \cdot -2)^8$	
$(-3 \cdot -2)^8$ 6^8 1,679,616	$(-3 \cdot -2)^8$ $(-3)^8 \cdot (-2)^8$ $6,561 \cdot 256$ 1,679,616

4. Aiyanna says that she can evaluate $(10 + (-2))^3$ using the power of a product exponent law. Explain to Aiyanna if she is correct.
- Aiyanna cannot apply the power of a product exponent law because the bases are being added not multiplied.*
5. Complete the statement about the power of a product exponent law.
- When a and b are nonzero real numbers, and m is an integer, the exponential expression $(a \cdot b)^m$ is equal to

- ☐ a^{bm} .
☒ $a^m \cdot b^m$.
☐ $a^m + b^m$.



Consider narrating the activity instructions for question 3 prior to releasing students to work together to complete. Students may skim the directions and miss the instructions on what should be done on each side of the table. If students do not do both methods, they will not make the connections that both methods result in the same answer, removing the notion that there is “one way” that works.



Consider providing liens of questioning to position students to make explicit connections between their responses and their analysis.

- How can you verify that Aiyanna is not correct?
- Why wouldn't an addition version work?
- Using the method from the table, how would you know if Aiyanna is correct or not?

3.1.4 Exploration: Power of a Power Exponent Law

Component Overview: Students will work collaboratively to discover the product of a power exponent law.



3.1.4 Exploration: Power of a Power Exponent Law

1. With your partner, complete the table.
 - a. Write each power of a power as a repeated multiplication expression using the outer exponent. Then, using the Product of Powers Exponent Law, rewrite the multiplication expression in the form a^m .

Power of a Power	Repeated Multiplication	Power of the Form a^m
$(4^2)^3$	$4^2 \cdot 4^2 \cdot 4^2$	4^6
$(10^4)^4$	$10^4 \cdot 10^4 \cdot 10^4 \cdot 10^4$	10^{16}
$(3^1)^5$	$3^1 \cdot 3^1 \cdot 3^1 \cdot 3^1 \cdot 3^1$	3^5

- b. How do the exponents in the first column relate to the exponents in the last column?
The exponent in the last column is the product of the exponents in the first column.
2. Two different methods for evaluating expressions with exponents are illustrated in the table. With your partner, follow the steps.
 - Select which column you will start in, and which column your partner will start in.
 - Review Method 1 and Method 2. When you are working in each column, make sure to follow the same process as in the example for that column.
 - After finding the value of each expression, verify your answers are the same. If not, find the error(s).
 - After each problem, swap columns so that you and your partner are practicing both methods.



Watch for misconceptions when students are working together on the table in question 1a. Students often forget the exponent that is attached to the base when showing their repeated multiplication.



Consider going over the instructions for question 2 with students to ensure students understand each partner's role in the activity.

3.1.4 Exploration: Power of a Power Exponent Law (continued)

Example	Method 1	Method 2
a.	$[(-2)^4]^2$	
	$[(-2)^4]^2$ $(16)^2$ $16 \cdot 16$ 256	$[(-2)^4]^2$ $(-2)^{4 \cdot 2}$ $(-2)^8$ 256
b.	$[(-6)^1]^8$	
	$[(-6)^1]^8$ $(-6)^8$ $(-6) \cdot (-6) \cdot (-6) \cdot (-6)$ $\cdot (-6) \cdot (-6)$ $\cdot (-6) \cdot (-6)$ 1,679,616	$[(-6)^1]^8$ $(-6)^{1 \times 8}$ $(-6)^8$ 1,679,616
c.	$[(-10)^2]^6$	
	$[(-10)^2]^6$ 100^6 $100 \cdot 100 \cdot 100 \cdot 100 \cdot 100$ $\cdot 100$ 1,000,000,000,000	$[(-10)^2]^6$ $(-10)^{2 \times 6}$ $(-10)^{12}$ 1,000,000,000,000



There are several possible misconceptions that can occur during this activity. Look for the following:

- Students may mistake the brackets for absolute value symbols.
- Students may forget to keep the negative symbol in their process.
- Students may add the exponents together because they think it looks like what they did in the prior activity.

3. Complete the statement.

When a is a nonzero real number, and m and n are integers, the exponential expression $(a^m)^n$ is equal to

- to
- ☐ a^{m+n} .
 - ☐ $a^m \cdot a^m$.
 - ☐ $a^{m \times n}$.

Power of a Power Exponent Law



When raising a base with a power to another power, multiply the exponents and keep the same base.



Allow students the time and opportunity to add the formal information for the power of a power exponent law to their note-taking guide, foldable, or additional Frayer Model to maintain their working reference guide.

3.1.5 Exploration: Identity Exponent Law

Component Overview: Students will work collaboratively to explore the identity exponent law by using repeated multiplication to discover the formal rule.



3.1.5 Exploration: Identity Exponent Law

1. With your partner, use the meaning of an exponent to complete the table. Write each product as a repeated multiplication expression, then write the value of each expression.

Exponential Expression	Repeated Multiplication	Value
10^1	10	10
100^1	100	100
11^1	11	11

2. Complete the statement about the identity exponent law.

When a is a real number, the exponential expression a^1

is equal to

- ☐ 1.
- ☐ $a + 1$.
- ☒ a .



Identity Exponent Law

When raising a base to the power of 1, the base is a factor one time, thus, has the same value as the base.



Some students may think that any number raised to the power of 1 will have a value of 1. To address this, draw emphasis to the repeated multiplication process that they have been doing throughout the lesson with other exponent laws.



Allow students the time and opportunity to add the formal information for the power of a power exponent law to their note-taking guide, foldable, or additional Frayer Model to maintain their working reference guide.

3.1.6 Your Turn!

Component Overview: Students will work independently to practice identifying and applying the laws of exponents learned in this lesson. Allow students to work on this independently and compare their answers with a partner.



3.1.6 Your Turn!

1. Complete the following steps for each expression.

- Rewrite each expression shown by using the laws of exponents.
- Choose the exponent law you used from the list.
- Evaluate the exponential expression.

$(2^3)^4$	☐ Product of powers law ☐ Power of product law ☐ Power of power law ☐ Identity exponent law
$(2^3)^4 = 2^{12} = 4,096$	

$0.5^4 \cdot 0.5^3$	☐ Product of powers law ☐ Power of product law ☐ Power of power law ☐ Identity exponent law
$0.5^4 \cdot 0.5^3 = 0.5^7 = 0.0078125 \approx 0.008$	

$\left(\frac{24}{8}\right)^1$	☐ Product of powers law ☐ Power of product law ☐ Power of power law ☐ Identity exponent law
$\left(\frac{24}{8}\right)^1 = \frac{24}{8} = 3$	

$(0.5 \cdot -8)^2$	☐ Product of powers law ☐ Power of product law ☐ Power of power law ☐ Identity exponent law
$(0.5 \cdot -8)^2 = 0.5^2 \cdot (-8)^2 = .25 \cdot 64 = 16$	



Some of the values when evaluating these problems can be tricky without a calculator. Consider allowing students to use a calculator during this activity.



When reviewing this activity, consider leveraging a polling protocol to increase student engagement. Put the exponent laws covered in this lesson in different corners of the room. Then display the problem and have students go to the corner that represents the exponent law they identified could be used. Have students discuss with one another and allow each group to share out. Allow the correct group at the end to share their evaluation of the expression. Repeat this for each problem.

3.1.7 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on identifying and applying laws of exponents. This activity should be done independently to provide accurate student-level data.



3.1.7 Wrap-Up: Ticket Out the Door

1. Rewrite the expressions using laws of exponents. State the exponent law used.

Expression	Exponent Law	Equivalent Expression
$12.75^5 \cdot 12.75^4$	<i>Product of Powers</i>	12.75^9
$\left[\left(\frac{1}{4}\right)^7\right]^2$	<i>Power of Powers</i>	$\left(\frac{1}{4}\right)^{14}$
$(12 \cdot 0.75)^3$	<i>Power of Product</i>	$12^3 \cdot 0.75^3$
285.6	<i>Identity Exponent</i>	285.6^1
$\left(\frac{-3}{5}\right)^1$	<i>Identity Exponent</i>	$\left(\frac{-3}{5}\right)$
$10^6 \cdot (-1.5)^6$	<i>Power of Product</i>	$(10 \cdot -1.5)^6$



The second column of the table has students identify the exponent law. If students get this wrong but the expression right, it can indicate a confusion issue with naming. If students get the exponent law right but the expression wrong, it could indicate a calculation error or switching the process with another law in error.